

Aaditty Tiwari

📍 San Jose, CA ✉️ atiwari306@gatech.edu ☎️ 650 931 7027 🌐 [AadiTiwari1](#)

Objective

Electrical Engineering Junior with first author research experience in Quantum Computing, Machine Learning, and Finite Element Analysis. Looking for an internship to apply and refine my skillset.

Education

Georgia Institute of Technology - B.S, Electrical Engineering. 3.80 GPA *Jan. 2026 – Dec. 2027*
University of California, Santa Cruz - Transferred out with a 3.95 GPA *June 2024 – Dec. 2025*

Research

Georgia Tech Quantum Computing Association ([Club website](#) [🔗](#)) *January 2026 – Present*
First Author, Club Officer

- Tasked with modeling a 3 qubit linear optical quantum computer in COMSOL to optimize design and minimize optical loss, enabling the club's goal of assembling a fully student-built quantum computer
- Modeled quantum state preparation section using polarizing beamsplitter, waveplates, and non polarizing beamsplitter with material-based phase calculations to initialize polarization to fiduciary state
- Presenting this work as a poster and paper at the COMSOL 2026 Boston Conference in October
- Wrote and presented grants for GT Student Foundation to purchase optical components, awarded \$10,000

University of California, Los Angeles ([TensorWorks Lab](#) [🔗](#)) *June 2023 – September 2025*
First Author

- Tasked with applying my experience with Quantum Machine Learning (QML) models to help a PhD Student with analyzing advantages/disadvantages of using QML in industrial turbine degradation forecasting
- Architected a model that uses a Quantum Long Short Term memory (QLSTM) as the Generator and classical Convolutional Neural Network as discriminator within a Generative Adversarial Network (GAN)
- Implemented Quantum GAN to forecast turbine's vibration signals, converging in 2.7x less epochs (49 classical epochs, 18 quantum) with only 169 QLSTM parameters VS 8,186,113 classical LSTM parameters.
- Deployed repository to a High Performance Computing cluster using SLURM to parameterize train/test split across 5 values and repeat each split 10 times to quantify variance in each model's RMSE performance
- [Paper](#) [🔗](#) accepted and presented to ESREL'24 conference Summer 2024. Currently revising conference paper for journal submission, code is open-sourced on [github](#) [🔗](#) with 14 stars and 5 forks

University of California, Santa Cruz ([Applied Optics Group](#) [🔗](#)) *November 2024 – December 2025*
Third Author

- Tasked with calculating electric field distribution within a solid-state nanopore in order to determine optimal structure to maximize nanopore chip's capture radius of particles
- Recreated exact nanopore chip structure used by PhD student in lab within COMSOL for finite element analysis of the electric field and voltage drop distribution
- Eliminated out of memory errors by creating custom mesh with varying element sizes depending on proximity to regions of interest and editing COMSOL's internal files to increase access to RAM
- Automated parameter sweep to vary the amount of nanopores (1,2,4,8,10) and distance between nanopores (100 nanometers to 6 micrometers in 100 nm increments) to calculate capture radius for each configuration
- Exported raw electric field distribution data from COMSOL to MATLAB to find an equation to express particle capture radius as a logistic function of nanopore count and placement

Skills, Certification, and Coursework

Languages & Frameworks: Python, Pytorch, Pennylane, Qiskit, Pandas, Numpy, MATLAB, C

Technologies: COMSOL, SLURM, VS Code, Cursor, Git, Supabase, Microsoft Office

Relevant Coursework: Embedded Systems with C, Digital Design & Assembly, GPU Programming, Opto-electronics & Photonics, Laser Theory & Applications, Modern Physics